

CSE 480

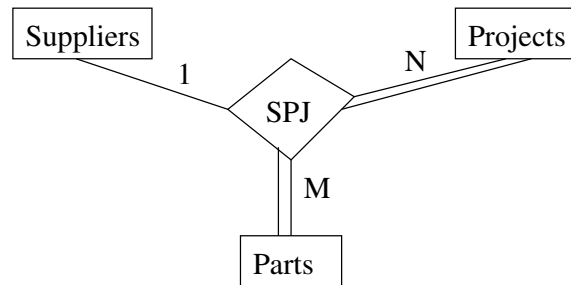
Homework 2: ER-EER Model- ANSWERS

DUE: Tuesday, February 4 (in class)

1. (10 points) Complete the ER diagram below with cardinality ratio and the participation constraint with the requirement that each (every) student must belong to only one department and a department may have multiple students.



2. (10 points) For the ternary relation below, give the statement describing the diagram. Your answer should capture the conceptual meaning and not a literal translation of the diagram.



A supplier can supply multiple parts to multiple projects. However, for a given project and a part there is at most one supplier.

3. (30 points) Draw an EER diagram for the following specifications. Your answer must include at least one weak entity type and one IS-A relation. Write down any assumptions you may make, as comments in the diagrams.

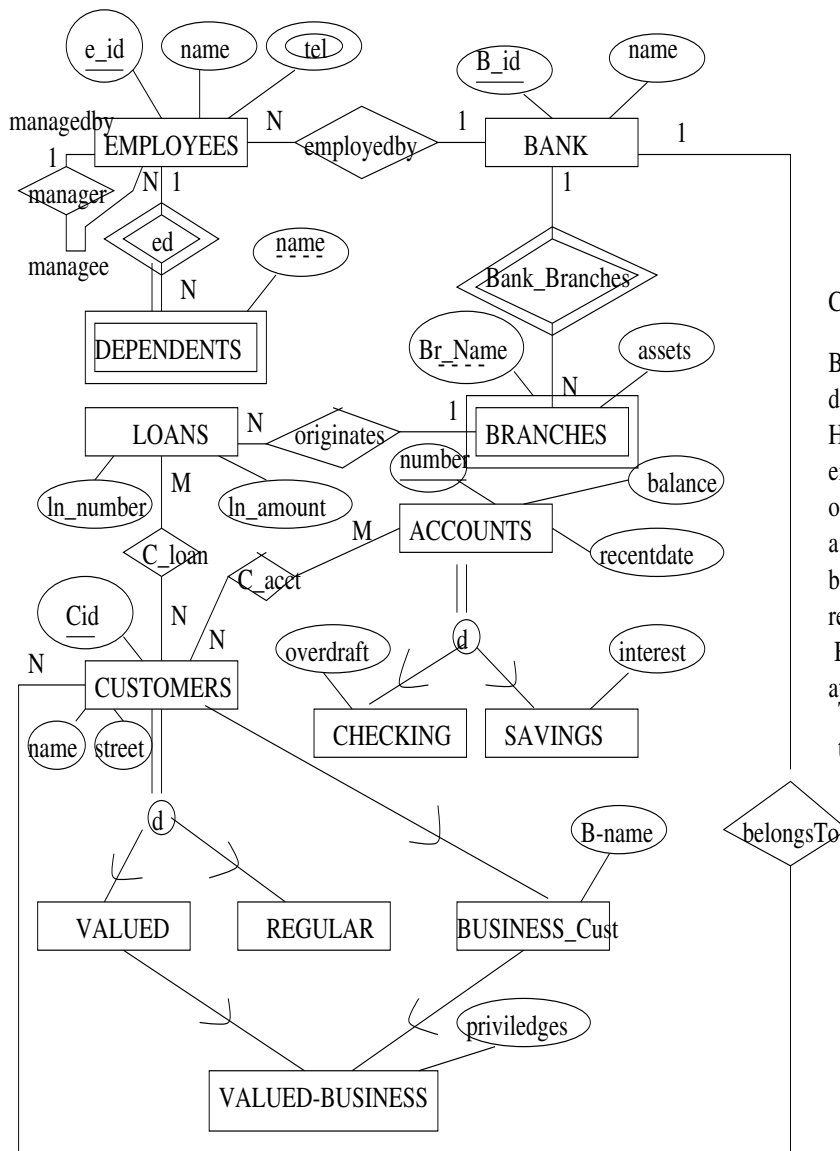
A bank is organized into branches. Each branch is located in a particular city and is identified by a unique name. The bank monitors the assets of each branch

Bank customers are identified by their customer-id values. There are three categories of customers, namely, valued-customers with a deposit of one million dollars or more, regular-customers with a deposit of less than one million dollars or business customers. We also keep track of business-customers who are valued-customers and privileges for valued-business-customers. The bank stores each customer's name, and the street and city where the customer lives. Customers may have accounts and can take out loans.

Bank employees are identified by their employee-id values and have dependents information. The bank administration stores the name and telephone number of each employee, the name of employee's dependents, and the employee-id number of the employee's manager. An employee can have multiple telephones. The bank also keeps track of the employee's start date and, thus, length of employment.

The bank offers two types of accounts - saving and checking accounts. Accounts can be held by more than one customer, and a customer can have more than one account. Each account is assigned a unique account number. The bank maintains a record of each account's balance, and the most recent date on which the account was accessed by each customer holding the account. In addition, each savings account has an interest rate, and overdrafts are recorded for each checking account.

A loan originates at a particular branch and can be held by one or more customers. A unique loan number identifies a loan. For each loan, the bank keeps track of the loan amount of the loan payments.



COMMENTS:

BRANCHES is totally dependent on BANK. However, BRANCHES can either be a composite attribute of BANK or it can be a weak entity type. However, because BRANCHES has a relationship type with LOANS, BRANCHES cannot be an attribute of BANK. Total dependence does not have to be a weak entity type.